# Evaluation runs: **sikv** (kv-snapshot-isolation-tla)

generated 2026-11-08 from brain-data 38584f536 2026-11-08 20:45:11 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

No run of the stateless engine on this benchmark.

## stateful

No run of the stateful engine on this benchmark.

## offline

### offline: pair **serializable**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 6 | conforms | 0.6 | | 10 min, 16 GB | 0.6 | 0.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 48 | 90 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 384 | 2520 | conforms | 0.5 | 8 | 10 min, 16 GB | 0.1 | 0.4 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 3840 | 113400 | conforms | 13.1 | 48 | 10 min, 16 GB | 0.3 | 12.8 | 0.3 | in-memory | brain05, 2026-11-02 |
| 6 | 46080 | 7484400 | conforms | 538.1 | 1138 | 10 min, 16 GB | 4.0 | 529.2 | 8.8 | in-memory | brain05, 2026-11-02 |

### offline: pair **si**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 6 | 1 | violation | 0.6 | 8 | 10 min, 16 GB | 0.6 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 48 | 90 | 4 | violation | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 384 | 2520 | 41 | violation | 0.5 | 8 | 10 min, 16 GB | 0.1 | 0.4 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 3840 | 113400 | 761 | violation | 11.8 | 34 | 10 min, 16 GB | 0.3 | 11.6 | 0.2 | in-memory | brain05, 2026-11-02 |
| 6 | 46080 | 7484400 | 20422 | violation | 493.6 | 685 | 10 min, 16 GB | 4.0 | 487.3 | 6.4 | in-memory | brain05, 2026-11-02 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 7 serializable fifo, 7 si fifo.

skipped: a smaller size had reached its limit: 7 serializable fifo, 7 si fifo, 8 serializable fifo, 8 si fifo, 9 si fifo, 9 serializable fifo, 10 serializable fifo, 10 si fifo.

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **serializable**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 6 | conforms | 3.3 | 169 | 10 min, 16 GB | 0.6 | 3.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 48 | 90 | conforms | 3.5 | 160 | 10 min, 16 GB | 0.1 | 3.5 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 384 | 2520 | conforms | 9.2 | 794 | 10 min, 16 GB | 0.1 | 9.2 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **si**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 6 | 1 | violation | 3.1 | 164 | 10 min, 16 GB | 0.6 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 48 | 90 | 4 | violation | 3.5 | 175 | 10 min, 16 GB | 0.1 | 3.5 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 384 | 2520 | 41 | violation | 9.2 | 746 | 10 min, 16 GB | 0.1 | 9.2 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Java implementation against the Rust specification).** planned but never scheduled (grid points beyond the ladder): 5 fifo si, 5 fifo serializable.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **serializable**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 6 | conforms | 3.1 | 160 | 10 min, 16 GB | 3.1 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 48 | 90 | conforms | 3.2 | 166 | 10 min, 16 GB | 3.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 384 | 2520 | conforms | 3.9 | 164 | 10 min, 16 GB | 3.8 | 0.4 | 0.1 | in-memory | brain01, 2026-11-02 |

### cross offline, Rust implementation against the Java specification: pair **si**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 6 | 1 | violation | 3.1 | 160 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 48 | 90 | 4 | violation | 3.2 | 166 | 10 min, 16 GB | 3.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 384 | 2520 | 41 | violation | 3.8 | 164 | 10 min, 16 GB | 3.8 | 0.4 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): 5 fifo si, 5 fifo serializable.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **serializable**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 6 | conforms | 3.1 | 156 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 48 | 90 | conforms | 3.2 | 162 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 384 | 2520 | conforms | 4.2 | 160 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **si**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 6 | 4 | violation | 3.1 | 156 | 1 h, 250 GB | 1 | brain01, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 48 | 90 | 48 | violation | 3.2 | 162 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 4 | 384 | 2520 | 1152 | violation | 4.2 | 160 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **serializable**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.4 | 169 | 10 min, 16 GB | 3.3 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 90 | 90 | conforms | 3.6 | 160 | 10 min, 16 GB | 3.5 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 2520 | 2520 | conforms | 9.2 | 794 | 10 min, 16 GB | 9.2 | 0.4 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Rust implementation against its Java twin: pair **si**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.1 | 164 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 90 | 90 | conforms | 3.5 | 175 | 10 min, 16 GB | 3.5 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 2520 | 2520 | conforms | 9.2 | 746 | 10 min, 16 GB | 9.2 | 0.4 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (twins offline, Rust implementation against its Java twin).** planned but never scheduled (grid points beyond the ladder): 5 fifo si, 5 fifo serializable.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **serializable**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.3 | 169 | 10 min, 16 GB | 0.3 | 3.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 90 | 90 | conforms | 3.5 | 160 | 10 min, 16 GB | 0.1 | 3.5 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 2520 | 2520 | conforms | 9.2 | 794 | 10 min, 16 GB | 0.4 | 9.2 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Java implementation against its Rust twin: pair **si**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.1 | 164 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 90 | 90 | conforms | 3.5 | 175 | 10 min, 16 GB | 0.1 | 3.5 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 2520 | 2520 | conforms | 9.2 | 746 | 10 min, 16 GB | 0.4 | 9.2 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (twins offline, Java implementation against its Rust twin).** planned but never scheduled (grid points beyond the ladder): 5 fifo si, 5 fifo serializable.

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **serializable**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.1 | 162 | 10 min, 16 GB | 3.1 | 0.7 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 90 | 90 | conforms | 3.8 | 168 | 10 min, 16 GB | 3.7 | 0.2 | 0.1 | in-memory | brain02, 2026-11-03 |
| 4 | 2520 | 2520 | conforms | 9.1 | 778 | 10 min, 16 GB | 9.1 | 0.5 | 0.0 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair si, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.6 | 172 | 10 min, 16 GB | 3.6 | 0.3 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 90 | 90 | conforms | 3.9 | 163 | 10 min, 16 GB | 3.9 | 0.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 2520 | 2520 | conforms | 10.1 | 862 | 10 min, 16 GB | 9.9 | 0.4 | 0.1 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** planned but never scheduled (grid points beyond the ladder): 5 fifo si, 5 fifo serializable.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair serializable, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.1 | 162 | 10 min, 16 GB | 0.7 | 3.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 90 | 90 | conforms | 3.8 | 168 | 10 min, 16 GB | 0.2 | 3.7 | 0.1 | in-memory | brain02, 2026-11-03 |
| 4 | 2520 | 2520 | conforms | 9.1 | 778 | 10 min, 16 GB | 0.5 | 9.1 | 0.0 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair si, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.6 | 172 | 10 min, 16 GB | 0.3 | 3.6 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 90 | 90 | conforms | 3.9 | 163 | 10 min, 16 GB | 0.1 | 3.9 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 2520 | 2520 | conforms | 10.0 | 862 | 10 min, 16 GB | 0.4 | 9.9 | 0.1 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** planned but never scheduled (grid points beyond the ladder): 5 fifo si, 5 fifo serializable.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair serializable, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 4.0 | 167 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 90 | 90 | conforms | 4.2 | 177 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 2520 | 2520 | conforms | 11.0 | 4789 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair si, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 4.9 | 162 | 1 h, 250 GB | | brain02, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 3 | 90 | 90 | conforms | 4.6 | 162 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 2520 | 2520 | conforms | 12.1 | 4779 | 1 h, 250 GB | | brain02, 2026-11-03 |

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **serializable**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.3 | 158 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 90 | 90 | conforms | 3.8 | 164 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 2520 | 2520 | conforms | 9.6 | 774 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **si**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 6 | conforms | 3.6 | 167 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 90 | 90 | conforms | 3.9 | 159 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 2520 | 2520 | conforms | 10.5 | 857 | 1 h, 250 GB | | brain02, 2026-11-03 |